

Economic Optimization of Predictive Maintenance for Industrial Assets: A Dynamic Decision-Making Framework

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Abstract

The transition to sustainable and efficient industrial operations requires a paradigm shift in asset management, moving beyond purely technical health monitoring to integrated economic decision-making. This paper presents a novel, patented methodology for the economic optimization of predictive maintenance. The proposed system continuously calculates the financial loss associated with asset performance degradation by comparing expected versus actual performance and scaling this difference by real-time market prices. A maintenance intervention is triggered only when the cumulative financial loss equals or exceeds the dynamic cost of maintenance. This approach ensures that every maintenance action is an economically profitable decision. We discuss the theoretical framework, mathematical modeling, and potential applications across diverse sectors, including renewable energy, demonstrating significant improvements in cumulative profitability compared to traditional maintenance strategies.

1. Introduction

In the context of global energy transitions and the pursuit of Net-Zero emissions, the operational efficiency of industrial assets—ranging from solar farms to complex geochemical systems like those explored for stimulated geological hydrogen production in Oman [1]—is paramount. Traditional Predictive Maintenance (PdM) systems have significantly advanced the ability to forecast equipment failures using sensor data and machine learning [2] [3]. However, these systems predominantly rely on technical thresholds (e.g., vibration limits, temperature anomalies) to trigger maintenance actions.

This technical-centric approach often neglects the dynamic economic environment in which these assets operate. For instance, performing maintenance during a period of high energy demand and peak market prices incurs a substantial opportunity cost, representing a form of “value leakage” [4]. Conversely, delaying maintenance when market prices are low might be economically optimal, even if technical degradation is occurring.

To address this critical gap, this paper introduces a patented decision-making system and method that integrates real-time economic variables with asset performance data to optimize maintenance scheduling [5].

2. Methodology: The Economic Decision Engine Framework

The core innovation of the proposed system is the continuous, real-time comparison between the economic loss of value due to performance degradation and the cost of a maintenance intervention.

2.1 Mathematical Model of Financial Loss

The system continuously monitors the performance of an asset using a network of IoT sensors. The financial loss (L_f) at any given time t is calculated based on the deviation of the measured performance (P_{actual}) from the expected optimal performance (P_{exp}).

This performance gap is then monetized by multiplying it by the real-time market price (M_p) of the asset's output (e.g., electricity feed-in tariff for a solar farm, or the cost of marine fuel for a ship experiencing hull biofouling).

The instantaneous financial loss is defined as: $\$L_{f, instantaneous}(t) = (P_{exp}(t) - P_{actual}(t)) \times M_p(t)\$$

The cumulative financial loss ($L_{f, cumulative}$) over a period from t_0 to t_n is the integral of the instantaneous loss: $\$L_{f, cumulative}(t_n) = \int_{t_0}^{t_n} [(P_{exp}(t) - P_{actual}(t)) \times M_p(t)] dt\$$

2.2 Dynamic Maintenance Cost

Simultaneously, the system calculates the cost of a maintenance operation (C_m). Crucially, C_m is not a static value but a dynamic variable that incorporates real-time factors such as labor costs, equipment availability, and the cost of consumables.

$$C_m(t) = f(\text{Labor}(t), \text{Parts}(t), \text{Logistics}(t))$$

2.3 The Decision Logic

The decision to authorize a maintenance operation is governed by a strict economic inequality. A maintenance command is triggered at time $t_{optimal}$ only when the cumulative financial loss reaches or exceeds the dynamic maintenance cost:

$$\text{Trigger Maintenance if: } L_{f, cumulative}(t) \geq C_m(t)$$

This logic ensures that the cost of intervention is fully justified by the accumulated loss of value, guaranteeing an economically profitable decision.

3. Results and Discussion

3.1 Dynamic Triggering Mechanism

The application of the proposed methodology fundamentally alters maintenance scheduling. **Figure 1** illustrates the dynamic interaction between cumulative financial loss and maintenance cost. Unlike traditional systems that trigger based on a fixed technical degradation line, the PREDAIOT system waits until the economic threshold is crossed. The dynamic nature of $C_m(t)$ means the optimal trigger point adapts to changing operational costs.

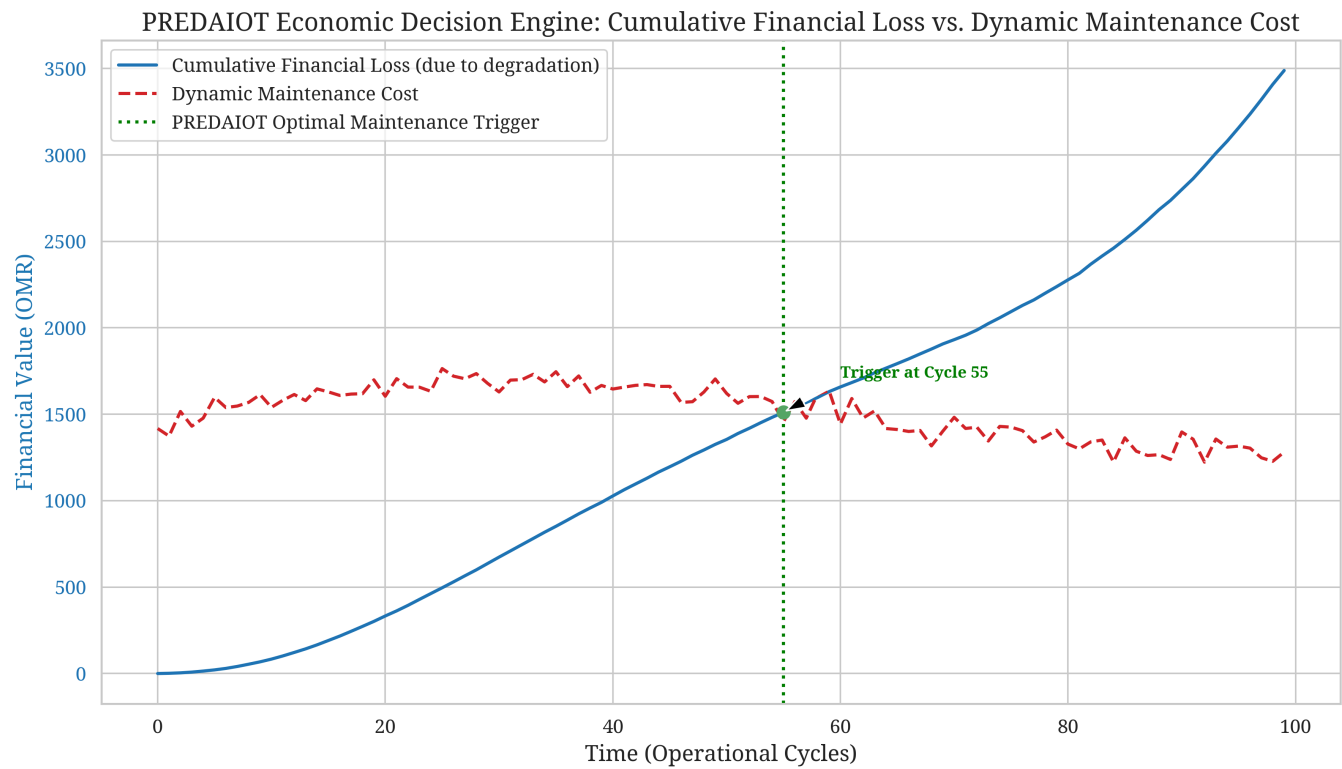


Figure 1: The decision logic of the patented system. Maintenance is triggered precisely when the cumulative financial loss (blue line) intersects the dynamic maintenance cost (red dashed line).

3.2 Cumulative Economic Impact

By ensuring that maintenance is only performed when economically viable, and by avoiding downtime during high-value periods, the system significantly enhances long-term profitability. **Figure 2** demonstrates a simulated comparison over a 365-day operational cycle. The

PREDAIOT Economic Decision Engine (EDE) strategy consistently outperforms traditional maintenance approaches, recovering substantial value leakage and expanding operational margins.

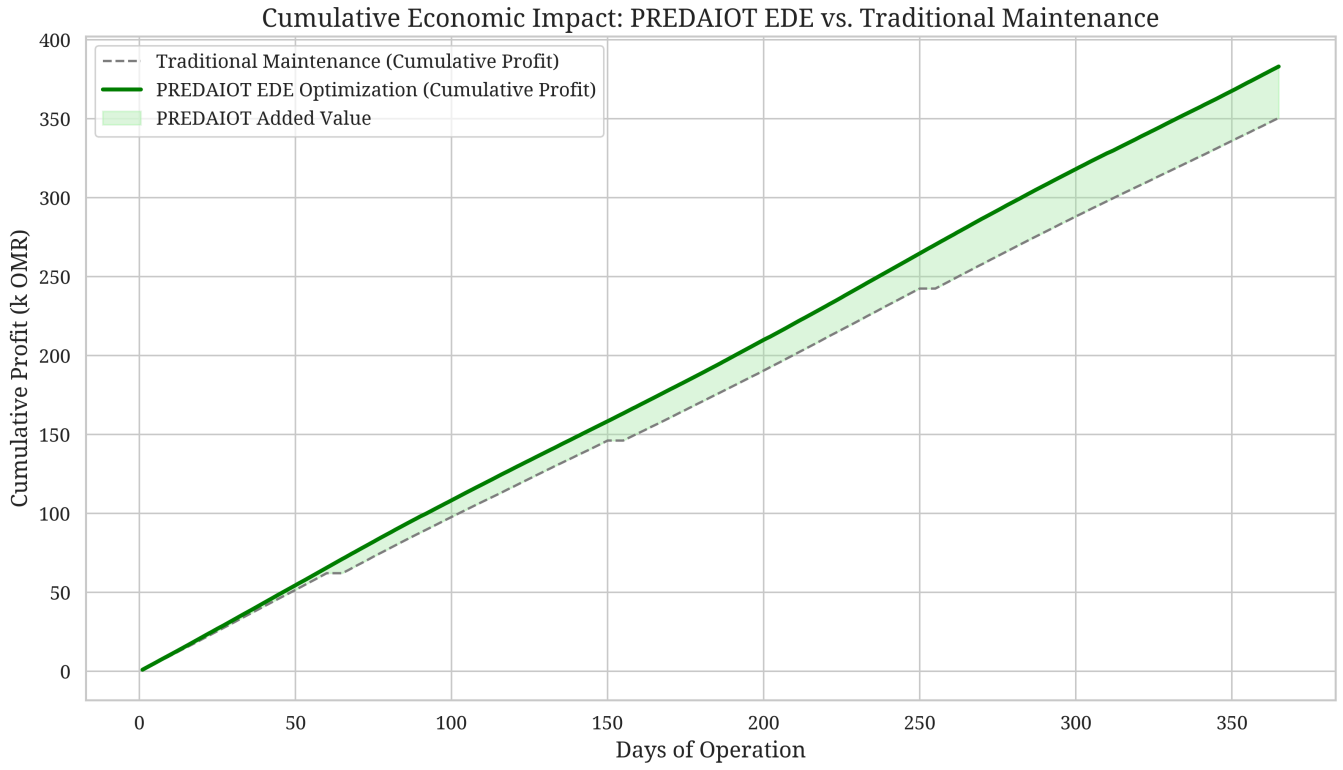


Figure 2: Cumulative profit comparison demonstrating the added value of the PREDAIOT EDE strategy over traditional maintenance methods.

3.3 Broad Applicability

While the mathematical model is universally applicable, the specific variables adapt to the asset type. As detailed in the patent claims [5]:

- **Solar Farms:** P_{actual} is electrical power; M_p is the electricity tariff.
- **Maritime Transport:** Performance loss is increased fuel consumption due to biofouling; M_p is the cost of marine fuel.
- **Aviation:** Performance loss is increased kerosene consumption; M_p is the cost of aviation fuel.

Furthermore, this economic optimization framework is highly relevant to emerging energy sectors, such as the stimulated geological hydrogen production systems currently being researched in the Samail Ophiolite in Oman [1]. Optimizing the maintenance of extraction and processing infrastructure in such novel environments will be critical for their commercial viability.

4. Conclusion

The integration of real-time economic data with predictive maintenance algorithms represents a necessary evolution in asset management. The patented methodology presented in this paper provides a robust, mathematically sound framework for ensuring that maintenance decisions are driven by profitability rather than purely technical thresholds. By continuously balancing financial loss against dynamic intervention costs, the system maximizes the economic value of industrial assets, offering a critical tool for industries navigating the complexities of the modern energy transition.

References

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